Lab 7: Fuzzing for Input Validation Bugs

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Github repo:
https://github.com/PrateekYV/L6_L7_55_car_insurance_premium.git

Screenshot 1: Normal and crashing runs

With valid input the program works. With an empty date of birth it crashes with an **IndexError** because driver_age assumes three '/'-separated parts.

```
D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7>python car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 ab 1234
Normalized registration: KA01AB1234

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000
Total claimed: 12000

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): 14/08/1999
Driver age in 2026: 27

D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7>python car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 ab 1234
Normalized registration: KA01AB1234

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000
Total claimed: 12000

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999):
Traceback (most recent call last):
  File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 39, in <module>
    main()
    ~~~~^^
  File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 36, in main
    print("Driver age in 2026:", driver_age(value))
                                 ~~~~~~~~~~^^^^^^^
  File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 23, in driver_age
    return current_year - int(dob.split('/')[2])
                          ~~~~~~~~~~~~~~~~~~^^^
IndexError: list index out of range

D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7>python car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 ab 1234
Normalized registration: KA01AB1234

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000;;abc
Traceback (most recent call last):
  File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 39, in <module>
    main()
    ~~~~^^
  File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 32, in main
    print("Total claimed:", total_claim_amount(value))
```

Entering the claims 12000;;abc makes total_claim_amount raise a **ValueError**, because int() cannot convert the empty or non-numeric entry.

```
D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7>python car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 ab 1234
Normalized registration: KA01AB1234

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000;;abc
Traceback (most recent call last):
  File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 39, in <module>
    main()
    ~~~~^^
  File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 32, in main
    print("Total claimed:", total_claim_amount(value))
                            ~~~~~~~~~~~~~~~~~~^^^^^^^
  File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 16, in total_claim_amount
    return sum(int(amount) for amount in text.split(';'))
  File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 16, in <genexpr>
    return sum(int(amount) for amount in text.split(';'))
               ~~~^^^^^^^^
ValueError: invalid literal for int() with base 10: ''

D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7>
```

Screenshot 2: Tests failing on the buggy module

On the original module, 7 of the 8 tests fail. Hypothesis reports minimal failing inputs such as None and the empty string.

```
D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7>python -m pytest test_car_insurance_premium_parser.py
============================= test session starts =============================
platform win32 -- Python 3.14.7, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7
plugins: anyio-4.15.1, hypothesis-6.170.14
collected 8 items

test_car_insurance_premium_parser.py FFFFFF.F                            [100%]

================================== FAILURES ===================================
_____________________ test_normalize_registration_no_crash _____________________

s = None

    @given(st.text() | st.none())
    def test_normalize_registration_no_crash(s):
        try:
>           normalize_registration(s)

test_car_insurance_premium_parser.py:13:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

plate = None

    def normalize_registration(plate):
        """
        Removes spaces and dashes from a registration number and upper-cases it.
        Assumes plate is a non-empty string.
        """
>       plate = plate.strip().upper()
                ^^^^^^^^^^^^

E       AttributeError: 'NoneType' object has no attribute 'strip'

car_insurance_premium_parser.py:6: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_normalize_registration_no_crash(s):
                    ^^^

test_car_insurance_premium_parser.py:11:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = None
```

```
_____________________ test_driver_age_result_is_none_or_sane_int _____________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 71, in test_driver_age_result_is_none_or_sane_int
  |     def test_driver_age_result_is_none_or_sane_int(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 72, in test_driver_age_result_is_none_or_sane_int
    |     result = driver_age(s)
    |   File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ~~~~~~~~~~~~~~~~~~^^^
    | IndexError: list index out of range
    | Failing test case: test_driver_age_result_is_none_or_sane_int(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\test_car_insurance_premium_parser.py", line 72, in test_driver_age_result_is_none_or_sane_int
    |     result = driver_age(s)
    |   File "D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7\car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_driver_age_result_is_none_or_sane_int(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
=========================== short test summary info ===========================
FAILED test_car_insurance_premium_parser.py::test_normalize_registration_no_crash - AssertionError: normalize_registration crashed with: AttributeError("'NoneType' object has no attribute
'strip'") on input None
FAILED test_car_insurance_premium_parser.py::test_total_claim_amount_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_car_insurance_premium_parser.py::test_driver_age_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_car_insurance_premium_parser.py::test_normalize_registration_output_is_clean - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_car_insurance_premium_parser.py::test_total_claim_amount_matches_sum_for_valid_input - ValueError: invalid literal for int() with base 10: ''
FAILED test_car_insurance_premium_parser.py::test_total_claim_amount_never_negative - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_car_insurance_premium_parser.py::test_driver_age_result_is_none_or_sane_int - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
========================= 7 failed, 1 passed in 12.18s ========================

D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7>
```

Screenshot 3: Fixed application output

The same kinds of bad input that crashed the original are now handled: the claims 12000;;abc;4500 total 16500 and the impossible date 31/02/1999 is reported as invalid.

```
D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7>python car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 ab 1234
Normalized registration: KA01AB1234

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000;;abc;4500
Total claimed: 16500

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): 31/02/1999
Invalid date of birth.

D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7>
```

Screenshot 4: All tests passing

After the fixes, all 8 Hypothesis tests pass, so none of the generated inputs crash the module.

```
D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7>python -m pytest test_car_insurance_premium_parser.py
============================= test session starts ==============================
platform win32 -- Python 3.14.7, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7
plugins: anyio-4.15.1, hypothesis-6.170.14
collected 8 items

test_car_insurance_premium_parser.py ........                            [100%]

============================== 8 passed in 2.71s ==============================

D:\pes\sem 5\SE\LAB\SE-Lab-Submissions\L6_L7_55_car_insurance_premium\LAB7>
```